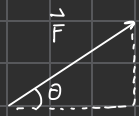


1. Forces in Equilibrium:



$$\vec{F} = \begin{pmatrix} \vec{F}\cos\theta \\ \vec{F}\sin\theta \end{pmatrix}$$

This is a combination of components.

If $R = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow |R| = \sqrt{x^2+y^2} = \sqrt{R \cdot R}$ Magnitude

$\Rightarrow \tan\left(\frac{y}{x}\right)$ Direction

R is defined as the resultant of forces X & Y.

$$\Sigma F = 0 \Leftrightarrow \begin{pmatrix} F_1\cos\theta \\ F_1\sin\theta \end{pmatrix} + \begin{pmatrix} F_2\cos\theta \\ F_2\sin\theta \end{pmatrix} + \begin{pmatrix} F_3\cos\theta \\ F_3\sin\theta \end{pmatrix} = 0$$

When forces are in equilibrium, $\Sigma F = 0$, hence resolve:

⇒ Horizontal & Vertical

⇒ Parallel & Perpendicular (to a slope)

Use $\Sigma \vec{F} = m\vec{a}$ to find direction of acceleration.

Resultants (ΣF) can be found by resolving forces and performing: $\Sigma F = \sum_{n=1}^{x} \begin{pmatrix} F_n\cos\theta_n \\ F_n\sin\theta_n \end{pmatrix}$, where θ is the angle between a surface & the angle

$F = ma \Leftrightarrow \vec{F} = m\vec{a} \Leftrightarrow \begin{pmatrix} F_1 \\ F_2 \end{pmatrix} = m\begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$